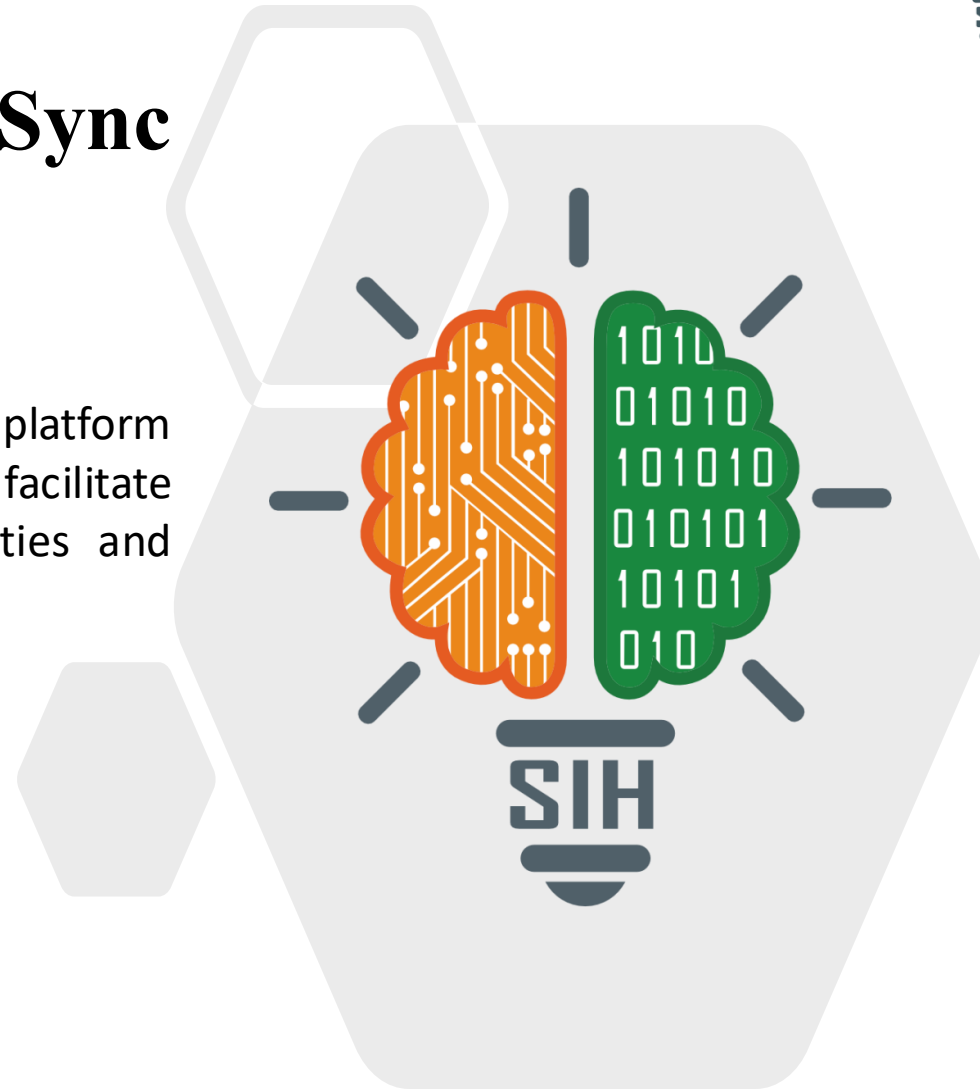




CivicSync

- **Problem Statement ID – 26043**
- **Problem Statement Title-** A digital platform to crowdsource societal challenges and facilitate collaborative problem solving through universities and industry partnerships
- **Theme- Smart Education**
- **PS Category- Software**
- **Team ID-**
- **Team Name- Mind Mavericks**



Problem

Proposed Solution

- ❖ **Problems remain unstructured** Citizens report real issues through different channels, but raw reports are not consistently converted into structured, actionable challenges.
- ❖ **Expertise remains disconnected** Universities, faculty, students, researchers and industry possess relevant capabilities, but finding the right solver for a specific societal problem is difficult.
- ❖ **Systemic problems are hidden** Similar complaints may come from multiple people or locations, but without clustering and prioritization, the larger underlying problem can remain fragmented.
- ❖ **The journey breaks after discovery** Problem reporting, solution development, implementation and impact measurement often exist as separate activities rather than one continuous process.
- ❖ **THE CORE GAP:** Problem → Expertise → Solution → Implementation → Impact
CivicSync bridges this gap through one connected intelligence and collaboration layer.

1. AI-Powered Challenge Intelligence

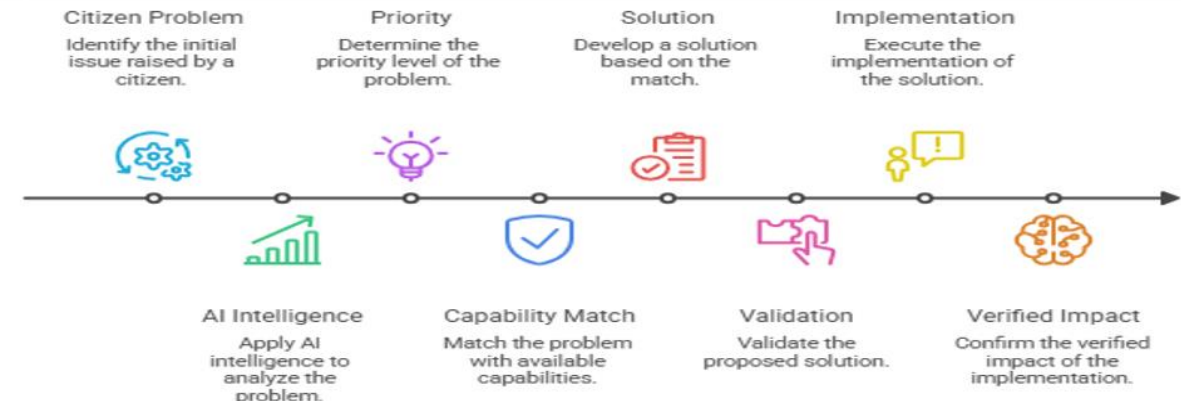
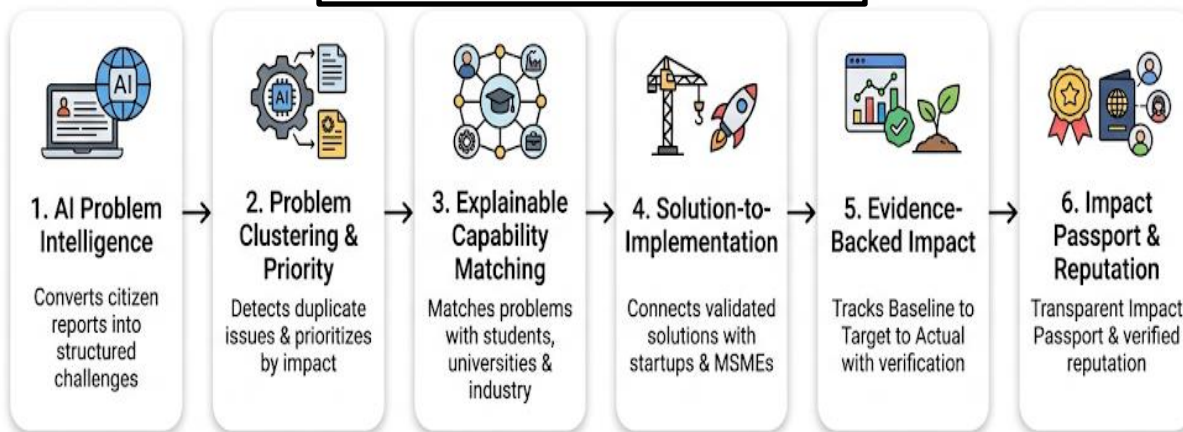
- ❖ **CivicSync** converts raw citizen reports into **structured, prioritized and actionable challenges** by identifying domain, severity, urgency and required expertise. Related problems can also be grouped into a larger challenge.

❖ Understand → Cluster → Prioritize

2. End-to-End Collaborative Problem Solving

- ❖ **CivicSync** connects challenges with **students, researchers, universities and industry**, enabling solution development, validation and implementation while tracking **Baseline → Target → Actual impact**.

Innovation & Uniqueness



CivicSync does not stop at finding a problem or an expert — it creates a traceable path from

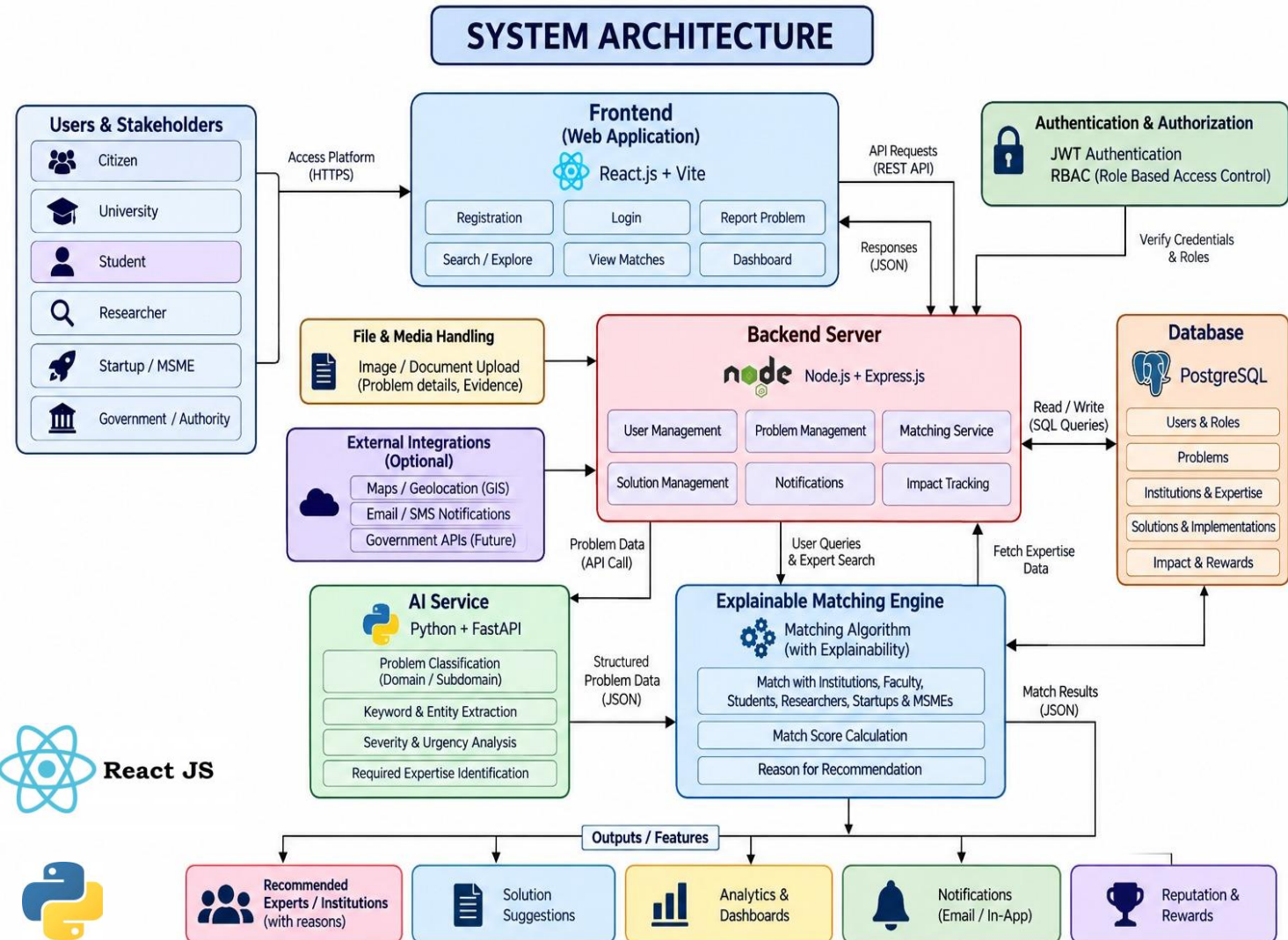
Problem → **Solution** → **Implementation** → **Verified Impact**

TECHNOLOGY STACK

- ❖ Frontend: HTML, Tailwind CSS, JavaScript, React.js
- ❖ Backend: Node.js, Express.js, REST APIs, bcrypt.
- ❖ RBAC(Role Based Access Control)
- ❖ Database: PostgreSQL
- ❖ Authentication: JWT Authentication
- ❖ AI/Intelligence: Python, FastAPI, Rule based NLP, Pattern Analysis, LLMs, Embeddings



SYSTEM ARCHITECTURE



FEATURES

1. **Citizen Problem Reporting :**

Citizens can submit real-world societal problems with location, category and affected population.

2. **AI Challenge Intelligence :**

AI identifies domain, subdomain, severity, urgency, keywords and required expertise from the problem.

3. **Explainable Expertise Matching :**

Matches problems with relevant institutions, faculty, students, researchers, startups and MSMEs with match scores and reasons.

4. **Duplicate & Similar Problem Detection :**

Identifies potentially duplicate or related problems to reduce repeated efforts.

5. **Implementation & Pilot Tracking :**

Tracks milestones, progress, evidence and blockers during solution implementation.

6. **Impact Measurement :**

Measures baseline vs actual outcomes, citizen feedback and verified impact.

7. **Reputation, Rankings & Rewards :**

Rewards verified meaningful contributions through reputation points, badges and rankings.

Technical Feasibility

- Modular stack: React + Node.js + FastAPI + PostgreSQL — UI, workflow, DB, and AI cleanly separated
- MVP-ready AI: rule-based NLP + pattern matching — no GPU or massive datasets needed

Operational Feasibility

- Human-in-the-loop: AI classifies & ranks; authorities validate before action
- Multi-role RBAC: tailored dashboards for Citizens, Students, Universities, MSMEs, Govt

Data Feasibility

- Cold-start ready: bootstrapped with curated capability profiles for institutions, experts, startups
- Built for growth: structured schema, ready for future vector/LLM integration

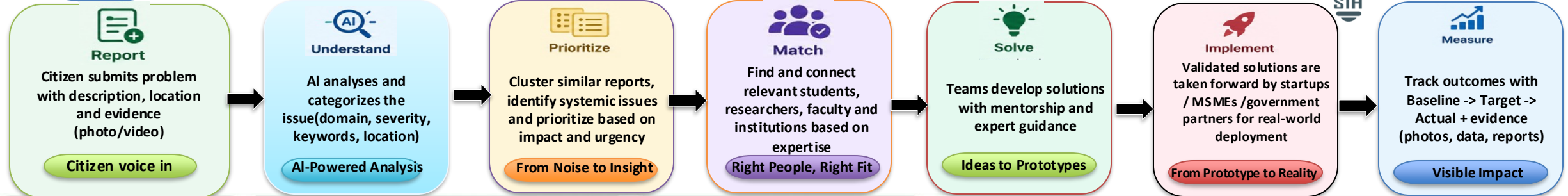
Scalability & Sustainability

- Phased rollout: District → State → National deployment
- Plug-and-play growth: new civic domains, AI models, GIS layers, and external registries

VIABILITY FORMULA Buildable Now → Low-Cost MVP → Human-Controlled → Modular Ecosystem → Scalable Architecture

HONEST LIMITATION MVP demonstrates rule-based AI with curated capability profiles — semantic vector search, fine-tuned LLMs, and real-time civic-portal API integration are scoped as Phase 2 extensions.

Challenges	Mitigation Strategy
AI Classification Errors	Confidence thresholding + mandatory human sign-off
Cold Start (Low Data)	Curated seed profiles + progressive self-onboarding
Duplicate / Spam Reports	Similarity clustering + geo-fencing + community moderation
Popularity / Vote Bias	Multi-factor score: severity + urgency + evidence + community signal
Privacy & Data Security	Consent + Role-Based Access Control (RBAC) + minimal PII storage

**1. PROBLEM SCALE**

CPGRAMS grievances received
(Central Ministries + States/UTs, June 2025)

2,06,323



Real citizen problems
across the country

2. EVEN 1% MAKES A DIFFERENCE

$$2,06,323 \times 1\% = 2,063$$

≈ **2,000+**
potential problem reports
can enter a structured
AI-assisted challenge →
solution pipeline

(Illustrative scenario, not a measured result)

3. FROM MANY REPORTS TO ONE SYSTEMIC CHALLENGE

50 related reports
(same or similar issue)



1 Master
Challenge

50 → 1 = **98%** fewer
fragmented challenge records
Better resource coordination and larger impact

IMPACT

80,000+
Grievances / Month

❖ 70,768 received and 63.135 redressed in June 2025.

❖ **Shows the need for better prioritization & follow-through.**

1.1 Cr+
Students Engaged

❖ Atal Innovation Mission reports 1.1 crore + students engaged through Atal Tinkering Labs.

❖ **A huge pool of young problem-solvers exists.**

3,500+
Startups Supported

❖ AIM reports 3,500+ startups supported through its incubation ecosystem.

❖ **Opportunity for a structured pathway from challenges to solution providers.**

66%
City Portals Enable Complaints

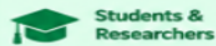
❖ UN E-Government Survey 2024: 66% of surveyed city portals offer complaint/feedback submission.

❖ **Reporting is digital; the next step is solving and tracking.**

Benefits for Key Stakeholders

Citizens

- Report local problems easily
- Track status and progress
- Get solutions from relevant experts



Students & Researchers

- Find real-world problems.
- Gain practical experience.
- Work on meaningful projects.
- Recognition opportunities



Startups & MSMEs

- Discover validated challenges
- Opportunities for pilot projects
- Access to academic expertise



Government / Authorities

- Prioritize problems effectively.
- Access wider solution ecosystem.
- Track implementation and outcomes.
- Data-driven policy planning

EXPECTED IMPACT

- Scattered citizen reports are converted into structured, prioritized challenges.
- Related reports are clustered into larger systemic challenges, helping focus resources on common problems.
- Relevant academic solutions can connect with startups/MSMEs for implementation.
- Projects move from Baseline → Target → Actual, supported by evidence and verification.

**WHERE THE GAP EXISTS**

Existing platforms solve important parts, but not the complete journey.

Capability	Existing Platforms (CPGRAMS / MyGov / AIM-ARISE)	Our Platform (CivicSync)
Citizen problem / grievance reporting	✓	✓
Structured problem intelligence	~	✓
Automated problem clustering	~	✓
Explainable expertise matching	—	✓
Implementation lifecycle tracking	~	✓
Baseline → Target → Actual	—	✓



= Core capability



= Limited / Not core capability



= Not a core capability

Expert Finding
Systems: A
Systematic Review

- ❖ **Evidence:** Review of 96 expert-finding studies; ~65% focused on academic environments. Expertise is commonly derived from profiles, documents and textual evidence.
- ❖ **CivicSync:** Capability-based matching of problems with relevant students, researchers, faculty and institutions.
- ❖ **Source:** <https://www.mdpi.com/2076-3417/9/20/4250>

Automated Citizen
Request
Classification

- ❖ **Evidence:** Research shows that large volumes of unstructured citizen requests can benefit from automated classification.
- ❖ **CivicSync:** AI-based problem intelligence for domain, severity, urgency and required expertise extraction.
- ❖ **Source:** <https://www.sciencedirect.com/science/article/pii/S0306457320309055>

Citizen Complaint
Pattern Analysis

- ❖ **Evidence:** Research on citizen complaints identifies spatial and temporal patterns that can reveal recurring or concentrated issues.
- ❖ **CivicSync:** Related-problem clustering and creation of systemic/master challenges.
- ❖ **Source:** <https://www.sciencedirect.com/science/article/pii/S2226585624000037>

Government/Ecosystem**CPGRAMS**
Government
of India

- ❖ Centralized public grievance reporting, tracking, feed backend appeal.
- pgportal.gov.in

MyGov
Government
of India

- ❖ Citizen engagement through tasks, discussions, polls, campaigns and submissions
- mygov.in

**Atal
Innovation
Mission
NITI-Aayog**

- ❖ Citizen engagement through tasks, discussions, polls, campaigns and submissions
- aim.gov.in

**Manthan —
Principal
Scientific
Adviser**

- ❖ Multi-stakeholder collaboration across government, academia & industry.
- psa.gov.in/manthan